
UCCI: Calibrated Uncertainty for Cost-Optimal LLM Cascade Routing

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Abstract

LLM cascades and model routing promise lower inference cost by sending easy queries to a small model and escalating hard ones to a large model, but most deployed routers use uncalibrated confidence scores and require per-workload threshold tuning. We present UCCI, a calibration-first router that maps token-level margin uncertainty to a per-query error probability via isotonic regression and selects the escalation threshold by constrained cost minimization. Under three explicit assumptions, threshold policies on the calibrated score are cost-optimal, and isotonic calibration achieves $O(n^{-1/3})$ sample complexity for expected calibration error (ECE). On a production named entity recognition workload of 75,000 queries served by 4B and 12B instruction-tuned LLMs on H100 GPUs, UCCI cuts inference cost by 31% (95% CI: [27%, 35%]) at micro-F1 = 0.91 while reducing ECE from 0.12 to 0.03. At the same operating point, UCCI beats entropy thresholding, split-conformal routing, and a FrugalGPT-style learned threshold. All cascade results use end-to-end routing on actual model outputs and measured H100 latency, not simulated routing from global accuracies or nominal API prices.

1. Introduction

Large language models perform well on structured prediction tasks but their per-query inference cost grows steeply with model size. For deployments where latency and serving cost matter, *cascaded inference*, serving most requests with a small model and escalating hard cases to a large one, is an attractive option (Chen et al., 2023a).

Cascade quality is bounded by the routing signal. LLM confidence scores, including raw token entropy and token margins, are typically miscalibrated and are sensitive to

prompt wording (Guo et al., 2017; Jiang et al., 2021). As a consequence, threshold choices that work in one workload often fail in another, and practitioners retune by hand for each deployment. The result is a gap between the theoretical promise of cascades and what production systems actually achieve.

This paper closes that gap by treating cascade routing as a calibrated decision problem rather than a tuning exercise. We make four claims, each supported by results in Section 6:

- **Formulation.** We pose cascade design as expected cost minimization subject to an accuracy constraint, and prove that threshold policies on calibrated error probabilities are cost-optimal under three explicit assumptions (Theorem 1).
- **Calibration.** We map token-margin uncertainty to error probability via isotonic regression, reducing expected calibration error from 0.12 to 0.03 on our workload.
- **End-to-end evaluation.** On a production named entity recognition (NER) workload with 75,000 labeled queries and 6 entity types, UCCI cuts cost by 31% (95% CI: [27%, 35%]) at micro-F1 = 0.91 versus large-model-only inference. Costs come from measured latency on H100 GPUs, and routing uses actual model outputs at test time, not simulated outcomes from global accuracies.
- **Comparison.** At the same F1 = 0.91 operating point, UCCI beats entropy thresholding (cost 2.31), conformal-prediction routing (2.18), and a FrugalGPT-style learned threshold (2.24).

Operationally, UCCI suggests calibrating first and thresholding second. A raw token-margin signal is weak when used directly, but after a small isotonic fit on a held-out calibration set it becomes a near-optimal routing score in our setting. Most of the cascade engineering value comes from making the routing score probabilistic, not from hand-tuning the final threshold.

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2. Related Work

Selective prediction and learning to defer. Selective classification (Geifman & El-Yaniv, 2017) and learning-to-defer (Madras et al., 2018; Mozannar & Sontag, 2020) study prediction with abstention or expert deferral under cost or risk constraints. These frameworks assume scalar confidence scores on classification heads. UCCI adapts the same idea to LLM structured outputs, where (i) uncertainty must be aggregated from token-level generation, (ii) raw token confidences are miscalibrated and require explicit correction, and (iii) production deployment makes end-to-end measurement, rather than simulated routing on global accuracies, the right validation target.

Uncertainty quantification and calibration. Calibration maps confidence scores to probabilities of correctness (Guo et al., 2017; Zadrozny & Elkan, 2002; Naeini et al., 2015; Platt, 1999). For LLMs, token probabilities are often miscalibrated (Jiang et al., 2021) and uncertainty estimation for free-form generation remains active (Kuhn et al., 2023; Farquhar et al., 2024). UCCI combines token-margin aggregation with isotonic regression to produce calibrated error probabilities (ECE = 0.03 on our workload) suitable for cost-constrained routing. Reliability layers for LLM structured extraction are studied in a complementary line of work (Kotte, 2026).

Conformal prediction for selective classification. Recent work applies conformal prediction to selective classification with coverage guarantees (Angelopoulos & Bates, 2021; Sadinle et al., 2019). We compare UCCI against a conformal-routing baseline that uses raw token-margin uncertainty as the nonconformity score.

LLM cascades and cost-aware routing. FrugalGPT (Chen et al., 2023a) routes between commercial APIs using learned confidence thresholds but does not impose calibration on the routing score. Hybrid LLM (Ding et al., 2024) learns a quality-aware router between a small and a large model, and RouteLLM (Ong et al., 2024) learns routers from preference data. RouterBench (Hu et al., 2024) standardizes evaluation for heterogeneous multi-LLM routing systems. Concurrent cascade work studies early abstention in settings where even the large model may decline to answer (Zellinger et al., 2025). Cascade ideas have also been applied to early exiting in language models (Schuster et al., 2022) and to multi-model speculative decoding (Chen et al., 2023b). Cost-aware deployment is a recurring concern in retrieval-augmented LLM systems for closed-domain applications (Sharma et al., 2024; Kotte et al., 2025). UCCI differs from these systems by making the routing score an explicitly calibrated error probability and by proving when a threshold on that probability is cost-optimal.

3. Problem Formulation

We consider inputs $x \sim \mathcal{D}$ for a structured prediction task with ground truth y . A small model f_s and a large model f_ℓ produce predictions $\hat{y}_s = f_s(x)$ and $\hat{y}_\ell = f_\ell(x)$. Let $\text{Acc}(\hat{y}, y) \in [0, 1]$ denote an accuracy metric (we use micro-averaged F1). Let c_s and c_ℓ denote per-query inference costs with $c_\ell > c_s$.

A routing policy $\pi : \mathcal{X} \rightarrow \{s, \ell\}$ selects which model to use:

$$\hat{y}_\pi(x) = \begin{cases} f_s(x) & \text{if } \pi(x) = s \\ f_\ell(x) & \text{if } \pi(x) = \ell \end{cases} \quad (1)$$

with cost $C_\pi(x) = c_s \mathbf{1}\{\pi(x) = s\} + c_\ell \mathbf{1}\{\pi(x) = \ell\}$ and accuracy $\text{Acc}_\pi(x) = \text{Acc}(\hat{y}_\pi(x), y)$.

Cost-optimal cascade design. Given a target accuracy $\tau \in [\alpha_s, \alpha_\ell]$ where $\alpha_s = \mathbb{E}[\text{Acc}(f_s(x), y)]$ and $\alpha_\ell = \mathbb{E}[\text{Acc}(f_\ell(x), y)]$, we seek

$$\min_{\pi} \mathbb{E}[C_\pi(x)] \quad (2)$$

$$\text{s.t. } \mathbb{E}[\text{Acc}_\pi(x)] \geq \tau. \quad (3)$$

4. Uncertainty-Calibrated Cascaded Inference

UCCI builds a routing policy in three steps: (i) extract a scalar uncertainty score from the small model’s generation, (ii) calibrate that score to an error probability, (iii) select a threshold by constrained cost minimization on a held-out validation set.

4.1. Token-level Uncertainty

For a generated sequence with tokens $t = 1, \dots, T$, let $p_{t,1}$ and $p_{t,2}$ denote the top-1 and top-2 next-token probabilities at position t under greedy decoding. Define the per-token margin $m_t = p_{t,1} - p_{t,2} \in [0, 1]$ and aggregate to

$$u(x) = 1 - \frac{1}{T} \sum_{t=1}^T m_t. \quad (4)$$

Larger $u(x)$ indicates greater uncertainty. We use greedy decoding throughout so that $p_{t,1}$ and $p_{t,2}$ are well defined per position.

4.2. Calibrating Uncertainty to Error Probability

For calibration, we use a binary correctness event: $e(x) = 1$ if $\hat{y}_s \neq y$ (treating the JSON output as exact-match across all entity fields), and $e(x) = 0$ otherwise. We report micro-F1 as the primary evaluation metric in Section 6; the binary event is used only to fit the calibration map.

The raw uncertainty $u(x)$ is not a probability. We learn a monotonic calibration map $g : [0, 1] \rightarrow [0, 1]$ such that

$$g(u) \approx \mathbb{P}(e(x) = 1 \mid u(x) = u). \quad (5)$$

We fit g by isotonic regression (Zadrozny & Elkan, 2002) on a calibration set $\mathcal{C} = \{(x_i, u_i, e_i)\}_{i=1}^n$ with $u_i = u(x_i)$ and $e_i = e(x_i)$.

4.3. Threshold Policy Selection

Given calibrated error probability $\hat{p}(x) = g(u(x))$, UCCI uses the threshold policy

$$\pi_\theta(x) = \begin{cases} s & \text{if } \hat{p}(x) \leq \theta \\ \ell & \text{if } \hat{p}(x) > \theta. \end{cases} \quad (6)$$

On a validation set $\mathcal{V}$ where both models have been run, we choose

$$\theta^* = \arg \min_{\theta} \widehat{\text{Cost}}(\pi_\theta) \quad \text{s.t.} \quad \widehat{\text{Acc}}(\pi_\theta) \geq \tau, \quad (7)$$

where $\widehat{\text{Cost}}(\pi_\theta) = |\mathcal{V}|^{-1} \sum_{x \in \mathcal{V}} C_{\pi_\theta}(x)$ uses actual costs and $\widehat{\text{Acc}}(\pi_\theta) = |\mathcal{V}|^{-1} \sum_{x \in \mathcal{V}} \text{Acc}_{\pi_\theta}(x)$ uses actual model outputs.

5. Theoretical Analysis

We give two results. The first establishes that threshold policies on calibrated error probability are cost-optimal under explicit assumptions; the second states a sample-complexity bound for the calibration step.

Theorem 1 (Optimality of threshold policies). *Assume (i) $c_\ell > c_s$, (ii) f_ℓ achieves a fixed accuracy α_ℓ on the test distribution that is invariant to which queries are escalated, and (iii) we have access to the calibrated error probability $\hat{p}(x) = \mathbb{P}(e(x) = 1 \mid u(x))$. Then among policies that depend only on $u(x)$, there exists a cost-optimal threshold policy π_θ for some $\theta \in [0, 1]$ that satisfies the accuracy constraint (3), and every cost-optimal policy agrees with such a π_θ up to tie-breaking on level sets of $\hat{p}$.*

Proof sketch. Escalating x replaces small-model accuracy $1 - \hat{p}(x)$ by the fixed accuracy α_ℓ at additional cost $c_\ell - c_s$. The marginal accuracy gain per unit cost is non-decreasing in $\hat{p}(x)$, so greedy escalation in decreasing $\hat{p}$ minimizes cost subject to the accuracy constraint. Since policies can depend only on $u(x)$ and g is non-decreasing in u , the greedy rule is a threshold on $\hat{p}$. Isotonic regression produces a piecewise-constant non-decreasing map, so $\hat{p}$ has atoms; tie-breaking among queries with equal $\hat{p}$ does not affect expected cost or accuracy, which gives the up-to-tie-breaking characterization. The full proof appears in Appendix A.1. $\square$

Proposition 2 (Sample complexity for calibration). *Under standard regularity conditions on the conditional error probability and bounded $u(x)$, isotonic regression on n calibration examples satisfies $\mathbb{E}[\text{ECE}(\hat{g})] = O(n^{-1/3})$.*

Proof sketch. Isotonic regression has a known $O(n^{-2/3})$ MSE rate against the true monotone conditional probability (Barlow et al., 1972). Cauchy-Schwarz converts an MSE bound into an L_1 (ECE) bound, giving $O(n^{-1/3})$. Concentration provides the high-probability statement. Details in Appendix A.2. $\square$

Assumption (ii) of Theorem 1 (large-model accuracy invariant to routing) is the most restrictive. If escalated queries are systematically harder, α_ℓ on the escalated subset can fall below the marginal α_ℓ . We test this assumption directly in Section 6.3 and discuss when it can fail in Section 7.

6. Experiments

6.1. Setup

Task and data. We evaluate on a NER workload from a production enterprise photo management system. Inputs are natural-language search queries; outputs are structured JSON with up to 6 entity types (camera, lens, aperture, shutter speed, ISO, focal length). The dataset contains 75,000 manually labeled queries. We split the dataset disjointly into calibration (22,500, 30%), validation (15,000, 20%), and test (37,500, 50%). Queries have mean length 6.2 words (median 5); 89% contain at least one entity, with mean 2.1 entities per query (median 2; max 4).

Models. f_s is a 4B-parameter instruction-tuned LLM; f_ℓ is a 12B-parameter instruction-tuned LLM. Both use identical prompts and greedy decoding, served via vLLM (Kwon et al., 2023) on H100 GPUs.

Measured costs. We measured end-to-end latency (prompt processing plus generation) on H100 GPUs over 100 queries with cold cache. Mean latency was 47.2 ms per query for the small model and 142.3 ms for the large model, giving a cost ratio $c_\ell/c_s = 3.02$. We report normalized costs with $c_s = 1.0$ and $c_\ell = 3.02$.

Cascade evaluation methodology. All cascade results below use end-to-end routing rather than simulated routing on global accuracies:

1. Fit the calibration map g on the calibration set.
2. Select the threshold θ^* on the validation set.
3. For each test query x , compute $\hat{p}(x) = g(u(x))$ and apply π_{θ^*} to choose the actual output of $f_s(x)$ or $f_\ell(x)$, accumulating the actual cost and accuracy.

This removes the selection-bias confound in evaluations that combine routing decisions with population-level accuracies: the reported numbers are what the deployed cascade would achieve on the test set.

Table 1. Single-model performance on the test set (37,500 queries).

Model	Micro F1	Cost
Small (4B)	0.847	1.00
Large (12B)	0.932	3.02

Baselines.

- *Always-small / Always-large.* Single-model inference.
- *Entropy threshold.* Route by uncalibrated mean token entropy.
- *Conformal prediction.* Apply split conformal prediction to the binary event “small model is correct,” using raw token-margin uncertainty $u(x)$ as the nonconformity score (not the calibrated $\hat{p}$, to keep the baseline independent). Choose threshold α^* on the validation set to control miscoverage and escalate queries with $u(x) > \alpha^*$, providing distribution-free coverage (Angelopoulos & Bates, 2021).
- *FrugalGPT-style.* Tune a confidence threshold on the validation set to meet the accuracy target (Chen et al., 2023a).

We do not run RouteLLM (Ong et al., 2024) or Hybrid LLM (Ding et al., 2024) as direct baselines because they require preference or quality-gap labels that are unavailable for this proprietary structured extraction workload. The FrugalGPT-style threshold is the closest implementable learning-based comparator from the available data: both models are run on the calibration and validation sets, and the router is selected to meet the same accuracy target as UCCI.

6.2. Results

Single-model baselines. Table 1 reports single-model accuracies on the test set. The 12B model is 8.5 micro-F1 points more accurate than the 4B model at $3\times$ the cost, establishing room for selective routing.

Calibration quality. Figure 1 shows the reliability diagram. Isotonic regression reduces ECE from 0.12 to 0.03 (95% CI: [0.02, 0.04] via bootstrap over the calibration set). The calibrated curve tracks the diagonal across the full $[0, 1]$ range.

Cost-accuracy trade-off. Figure 2 shows the Pareto frontier on the test set under end-to-end cascade evaluation. At micro-F1 = 0.91, UCCI achieves cost = 2.08 versus 3.02 for the large model alone, a 31% reduction (95% CI: [27%, 35%] via bootstrap over queries).

Table 2 compares all routing methods at the same F1 = 0.91 operating point and at a matched cost budget of 2.00. At

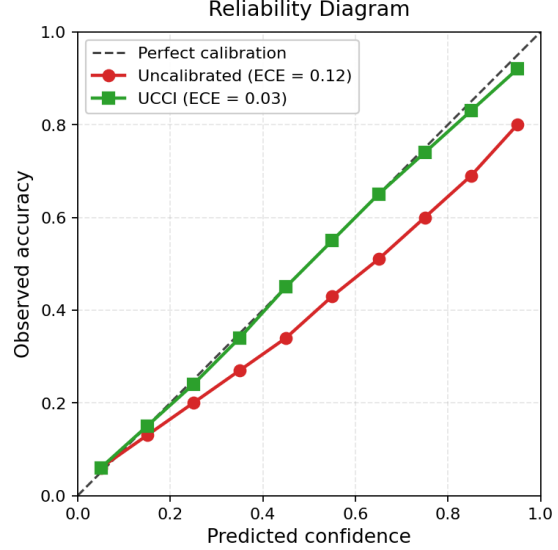


Figure 1. Reliability diagram on the calibration set. Isotonic regression reduces ECE from 0.12 (uncalibrated token margin) to 0.03 (UCCI).

Table 2. Cascade routing comparison on the test set. Top block: methods evaluated at the F1 = 0.91 operating point. Bottom block: methods evaluated at a matched cost budget of 2.00.

Method	Cost	Δ F1 vs target
UCCI (ours)	2.08	+0.00
Conformal prediction	2.18	+0.00
FrugalGPT-style	2.24	+0.00
Entropy threshold	2.31	+0.00
Large-only	3.02	+0.02
Small-only	1.00	−0.06
UCCI (ours)	2.00	+0.01
Entropy threshold	2.00	−0.02

the F1 target, UCCI achieves the lowest cost (2.08), beating entropy (2.31, 11% higher cost), conformal prediction (2.18, 5% higher), and FrugalGPT-style routing (2.24, 8% higher). At the matched cost budget, UCCI exceeds the F1 target by 0.01 while entropy thresholding falls 0.02 below. Small-only inference is included as a lower-bound anchor and falls 0.06 below the F1 target.

Table 3 reports the same UCCI routing decisions under hypothetical large/small cost ratios. Since every escalation has the same marginal cost, the accuracy-feasible threshold is unchanged; only the reported cost scale changes.

Per-entity heterogeneity. Table 4 shows that the small-large gap varies materially across entity types, from 1.8 F1 points on focal length to 14.3 points on lens. This heterogeneity is what makes adaptive routing valuable: there is no single global accuracy gap that a fixed-rate escalation can

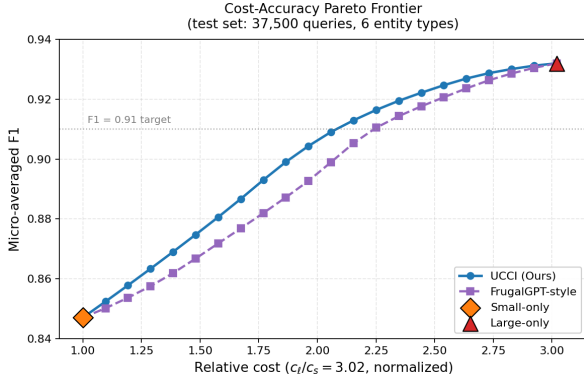


Figure 2. Cost-accuracy Pareto frontier on the test set. UCCI dominates the FrugalGPT-style and single-model baselines. Comparisons against entropy and conformal at matched operating points appear in Table 2. Every point uses end-to-end cascade evaluation with measured costs.

Table 3. Cost-ratio sensitivity for the UCCI routing policy at F1 = 0.91.

c_ℓ/c_s	UCCI cost	Savings vs large-only
3.02	2.08	31%
5.00	3.14	37%
10.00	5.81	42%

target.

6.3. Ablations and Diagnostics

Calibration method. Isotonic regression beats temperature scaling and uncalibrated routing under matched validation budget, reaching ECE = 0.03 versus 0.08 for temperature scaling on the same calibration set. Detailed numbers are in Appendix B.4.

Uncertainty signal. Among the three uncertainty signals we considered (token margin, predictive entropy, max probability), token margin gives the best cost-accuracy curve. The likely reason is that in structured JSON outputs the discriminating decisions are spread across a small number of high-information tokens, where a top-2 margin captures local ambiguity better than full entropy over the vocabulary. Semantic entropy is a strong recent signal for free-form generation (Farquhar et al., 2024); we do not use it here because structured JSON extraction concentrates the deciding evidence in a small number of schema tokens and because semantic-entropy estimation requires multiple samples or clustering of meanings, which changes the serving cost model.

Validating Theorem 1, assumption (ii). Theorem 1 assumes that f_ℓ achieves a fixed accuracy α_ℓ on the test distribution, regardless of which queries are escalated. We test

Table 4. Per-entity micro-F1 on the test set.

Entity type	Small	Large
Camera	0.71	0.89
Lens	0.68	0.86
Aperture	0.93	0.96
Shutter speed	0.91	0.95
ISO	0.92	0.95
Focal length	0.96	0.98
Overall (micro)	0.847	0.932

this directly: at the F1 = 0.91 operating point, f_ℓ achieves micro-F1 = 0.928 on escalated queries, versus 0.932 on the full test set, a difference of 0.004. The assumption holds approximately in our setting and the theoretical framework is applicable.

Falsification regime. The optimality argument relies on calibrated $\hat{p}$ being informative about the per-query error rate. This can fail when uncertainty is heavy-tailed: a small fraction of queries can sit in the body of the $u(x)$ distribution while having high true error probability, in which case threshold routing escalates fewer of them than an oracle would. We do not have a direct measurement of margin-tail behavior on this workload, but the largest residual gap between UCCI and the large model concentrates on the camera and lens entity types, where the small-large F1 gap is largest (Table 4). Appendix B.5 discusses this regime in more detail.

7. Discussion and Limitations

Scope. Our evaluation is on a single domain (photography NER) with 75,000 queries. The framework is task-agnostic but the absolute numbers (31% cost reduction, ECE = 0.03) are workload specific. Broader validation across tasks, languages, and model families would strengthen the generalization claim. We do not evaluate on RouterBench (Hu et al., 2024) because it targets API-level routing across heterogeneous tasks, while UCCI targets calibrated two-model cascades on a fixed structured-prediction workload with measured latency. A natural next step is to replicate the method on a public NER benchmark such as CoNLL-2003 and on a different small/large model family.

Theoretical assumption. Theorem 1 assumes the large model’s accuracy is invariant to routing. The empirical check in Section 6.3 shows a gap of only 0.004 on our workload, but this assumption can break in domains where the small model’s hard queries are also disproportionately hard for the large model. When that happens, the threshold policy is no longer guaranteed to be globally optimal; it remains a calibrated heuristic, and the empirical comparison

to entropy, conformal, and FrugalGPT-style baselines is unaffected.

Static calibration. UCCI fits the calibration map on a held-out batch. In streaming deployments with distribution shift, online or continual recalibration would be needed; the same calibrated threshold framework applies, but the fitting procedure must be adapted.

Cost model. We measure cost in H100 latency. Production deployments may instead optimize for dollar cost, throughput, energy, or end-user latency under batching. Each requires re-deriving c_s and c_ℓ , but does not change the calibration step or the optimality argument.

8. Conclusion

UCCI calibrates token-level uncertainty into error probabilities and selects an escalation threshold by constrained cost minimization. On a production NER workload of 75,000 queries, this reduces inference cost by 31% at micro-F1 = 0.91 versus large-model-only inference, and beats entropy, conformal, and FrugalGPT-style baselines at the same operating point. The empirical headline is that calibration, not threshold tuning, is the part of the pipeline worth engineering: a small isotonic fit converts a noisy token-margin signal into a routing score that is close to optimal in our setting.

Ethics Statement

Reducing inference cost can broaden access to LLM-based systems, and can also lower the barrier to deploying systems with potential for harm. Practitioners should consider the use case and the population of escalated queries before deploying calibrated cascades, particularly in high-stakes settings.

Reproducibility Statement

Hyperparameters, data splits, model configurations, and evaluation protocols are specified in Section 6.1 and Appendix B.2. Code and calibration scripts are released at <https://github.com/varunkotte6/ucci>. The raw production query data cannot be released due to privacy constraints; the repository includes the calibration, threshold-selection, and evaluation interfaces together with synthetic examples and per-split summary statistics sufficient to audit the pipeline.

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A. Full Proofs

A.1. Proof of Theorem 1

Proof. Let π be any policy that depends on $u(x)$ and satisfies constraint (3). For any input x , escalating to f_ℓ gives accuracy α_ℓ at cost c_ℓ , while keeping f_s gives expected accuracy $\mathbb{E}[\text{Acc}(f_s(x), y) \mid u(x)] = 1 - \hat{p}(x)$ at cost c_s .

The marginal accuracy gain from escalating x is

$$\Delta \text{Acc}(x) = \alpha_\ell - (1 - \hat{p}(x)) = \alpha_\ell - 1 + \hat{p}(x), \quad (8)$$

at marginal cost $\Delta c = c_\ell - c_s > 0$.

To minimize total cost subject to $\mathbb{E}[\text{Acc}_\pi(x)] \geq \tau$, we should escalate queries in decreasing order of $\Delta \text{Acc}(x)/\Delta c$, equivalently in decreasing order of $\hat{p}(x)$, until the constraint is met. This greedy allocation is implemented by choosing the smallest θ^* such that

$$\mathbb{E}[\text{Acc}_{\pi_{\theta^*}}(x)] \geq \tau, \quad (9)$$

escalating all x with $\hat{p}(x) > \theta^*$ and using f_s otherwise. Because policies can depend only on $u(x)$ and $\hat{p} = g(u)$ is non-decreasing, every cost-optimal policy reduces to a threshold on $\hat{p}(x)$.

When g is fit by isotonic regression, $\hat{p}$ is piecewise constant and therefore has atoms. On a level set $L_\theta = \{x : \hat{p}(x) = \theta\}$, every policy that escalates the same fraction of L_θ yields the same expected cost and accuracy, regardless of which specific queries in L_θ it picks. The threshold characterization is therefore unique up to this measure-zero tie-breaking on level sets of $\hat{p}$. $\square$

A.2. Proof of Proposition 2

Proof. Isotonic regression is a non-parametric maximum-likelihood estimator for monotonic functions. Under mild regularity conditions on the conditional probability and bounded $u(x)$, the mean squared error between the fitted $\hat{g}$ and the true conditional probability g^* satisfies

$$\mathbb{E}[\|\hat{g} - g^*\|_2^2] = O(n^{-2/3}), \quad (10)$$

for n calibration examples (Barlow et al., 1972). Expected calibration error is an L_1 distance between predicted and empirical probabilities, which by Cauchy-Schwarz satisfies

$$\text{ECE} \leq \sqrt{\mathbb{E}[\|\hat{g} - g^*\|_2^2]} = O(n^{-1/3}). \quad (11)$$

A standard concentration argument turns the expectation into a high-probability statement. $\square$

B. Additional Experimental Details

B.1. Prompt Format

All models use the following prompt template:

```
Extract entities from this photo search query.
Return JSON with fields: camera, lens,
aperture, shutter_speed, iso, focal_length.
```

```
Query: [USER_INPUT]
Output:
```

B.2. Hyperparameters

- Decoding temperature: 0 (greedy).
- Maximum generated tokens: 256.
- vLLM version: 0.4.2.

- GPU: NVIDIA H100 80GB.
- Calibration: isotonic regression as implemented in standard open-source libraries (default settings).

B.3. Cost Measurement Details

End-to-end latency was averaged over 100 queries with cold cache:

- Small model (4B): 47.2 ± 3.1 ms per query.
- Large model (12B): 142.3 ± 8.7 ms per query.
- Ratio: $c_\ell/c_s = 3.02$.

B.4. Calibration Method Ablation

We compared isotonic regression against temperature scaling (Guo et al., 2017) and uncalibrated routing under the same calibration set, validation set, and threshold-selection procedure. On the calibration set, isotonic regression achieved $\text{ECE} = 0.03$ versus $\text{ECE} = 0.08$ for temperature scaling. Uncalibrated routing has $\text{ECE} = 0.12$ (the raw token margin). Temperature scaling is constrained to a single-parameter monotone rescaling, which is too rigid to correct the non-monotone shape of token-margin miscalibration on this workload; isotonic regression has the right inductive bias.

B.5. Falsification Regime: Heavy-Tailed Margins

The optimality argument in Theorem 1 treats $\hat{p}(x)$ as informative about the per-query error rate. When token margins are heavy-tailed, a small fraction of queries can have $u(x)$ in the body of the distribution but very high true error probability; calibration on a finite sample cannot fully correct this, and threshold routing escalates fewer of these queries than an oracle policy would. We did not directly measure margin-tail statistics on our workload, so we do not claim a quantitative bound on the size of this effect. We note that the residual cost gap between UCCI and an oracle (large-only at target F1) is concentrated on the camera and lens entity types (Table 4), which is consistent with the heavy-tail hypothesis but does not establish it. On workloads where margins are heavier-tailed than ours, the relative gain of UCCI over entropy thresholding should persist, but the absolute cost reduction at a given accuracy target may shrink.